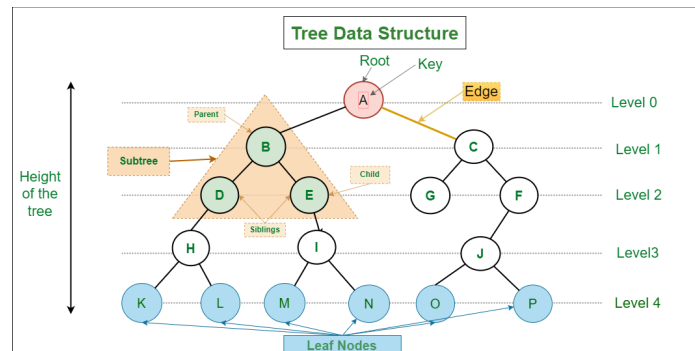


Terms:



1) **Root** - The root, is the *starting* position where the first node sits.

The root in the image about is {A}.

2) **Parent** - To easily explain this, **Root** = starting $\rightarrow \forall n, n = \text{node that come from the root, are know as } \textit{children}$ and root would be the **Parent**.

$B = \{D, E\}$ where D, E are children and B is the Parent

3) **Child** - Again, let's make it easy - above I basically explained it... but if you don't understand... the *child* in the case about would be the nodes descending from the **root**. (see above case)

4) **Sibling** - Those who are *children* to the same **Parent** are **Siblings**.

with the $B = \{D, E\}$ case - D, E are siblings! They both have B as a Parent.

5) **Descendents** - This is hard to explain without a visual but essentially if you pick a node within a tree - it's descendants are the nodes that are **Below** it.

$B = \{D, E, H, I, K, L, M, N\}$ are all descendents of B.

6) **Ancestor** - Ancestors are the nodes that are '*above*' the current node being looked at (which I will elaborate more on)

$L = \{H, D, B, A\}$ are all nodes that are Ancestors of L.

7) **Degree of node** - The degree is hard to explain without a visual - but the **Degree** is essentially the amount of

8) **Internal/External (Terminal/Non-Terminal)** - Nodes that leafs are **Non-Terminals** $\forall$ other nodes they are said to be **Terminals**.

9) **Levels** - Levels are the **depth** of a tree, starting at 1 index. (Height starts at 0)

10) **Height** - The height of the tree starting from 0 index, essentially when you draw it out you will assume the **root** to be the [0] index.

11) **Forest** - essentially a forrest is when you have a collection of trees.

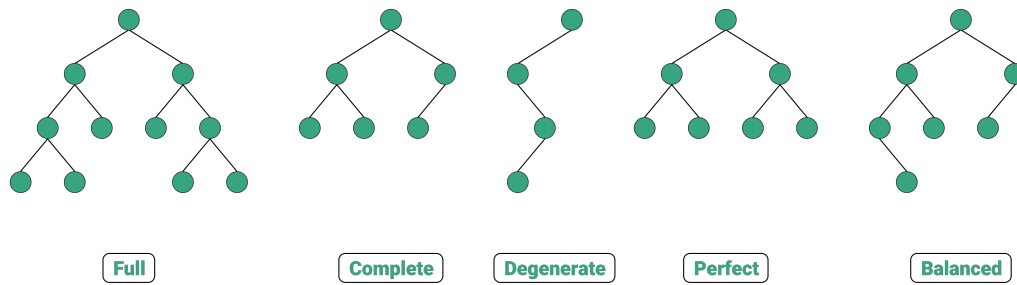


Figure 1: Tree Structures

- 1) **Full** - A full binary tree is also a **Strict Binary Tree**!
- 2) **Complete** - A tree that is filled with nodes from **Left to right** - or fills by lvl.
- 3) **Degenerate** - A tree that is Degenerate is a tree that
- 4) **Perfect** - A tree that is Perfect $\rightarrow$ the tree will have the **Maximum amount of nodes possible** $\rightarrow n = 2^{h+1} - 1$
This one is Also a **Strict Binary Tree** but also a **Complete Binary Tree**.
- 5) **Balanced** - A tree that is said to be **Balanced**

$$T(n) = \frac{(2*n)_c n}{n+1}$$